



WATER QUALITY MONITORING FOR HARARE BOREHOLES

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THE PROBLEM – HARARE, ZIMBABWE

- Due to the intermittent flow of other sources, many households (80%) rely on boreholes (private or communal).
- Poor sanitation and sewage management lead to borehole contamination.
- Residents paying roughly £20 a month for unreliable water.
- Current Approach: Mass chlorine dosing to kill bacteria without knowledge of the water's level of contamination.
 - Costly, Inefficient, Tastes and Smells Bad, Health Effects.



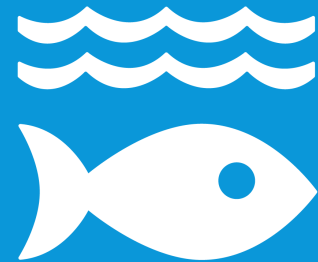
3 GOOD HEALTH
AND WELL-BEING



6 CLEAN WATER
AND SANITATION



14 LIFE
BELOW WATER



STAKEHOLDERS



Government:

Wants: Population well-being, economic growth, SDG development, and culture retention.

Limitations: Expenditures, lead times, and prioritisation.



Households:

Wants: Clean drinking water that is easy, cheap, and reliable to access.

Limitations: Income (2,400 USD/year), transportation, time, and technical fluency.



Healthcare System:

Wants: Healthy population, strong infrastructure/equipment, and highly educated personnel.

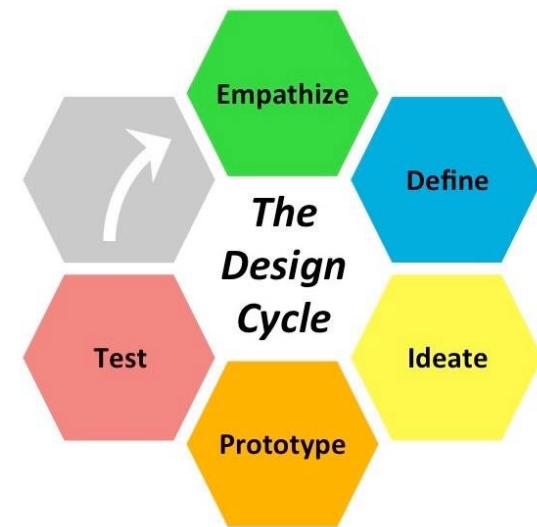
Limitations: Disease outbreaks via water sources, overcrowded system, and education limitations.

EXISTING SOLUTIONS

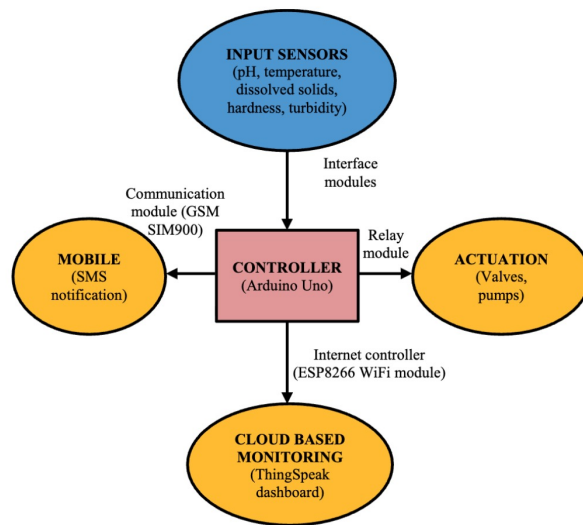
Golden Standard	Other Approaches	Our Approach
Microscopy – Staff, equipment & skill heavy <i>WaterScope</i> – Low-cost 3D-printed microscope. Cuts testing to 1–2hrs but still requires physical sample handling & incubation time. ↳ <i>Emerging competitor</i>	PCR – Highly accurate but expensive, slow lab method Chemical Test Kits + Spectrophotometry – 24–48hr results, proxy based. <i>All produce snapshots, not continuous data</i>	IoT Sensor Networks • Real-time, continuous monitoring • Multi-parameter • No lab required • Low cost • Cloud integration for instant alerts

PARTNERSHIP – AQUA SOLUTIONS

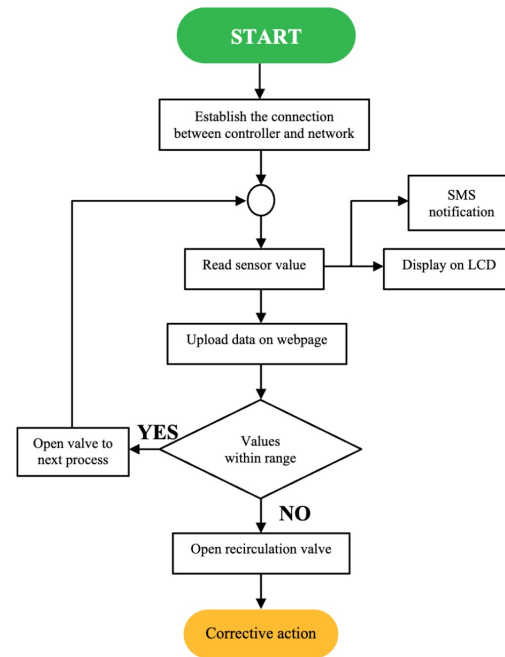
- 2022: Thesis for BSc in Industrial and Manufacturing Engineering
- Design Cycle:
 - **Prototyping:** 6 sensors, fuzzy logic → Successful!
 - **Trials:** System failure, sensor degradation.
 - **Feedback:** Consumers were excited by his idea and demonstrated WTP.
- Back to the drawing board, but having learned a lot along the way.



PREVIOUS DESIGN



Overview



Control Protocol

KEY CHALLENGES

- **Microbial Indicator:** Microbial sensors are only available in labs (costly, long lead times).
 - In the initial prototype, loosely correlated sensors were picked based off availability.
 - Can we find proxies indicative of bacteria presence? the biggest threat to human safety!
- **Sensor Vulnerabilities:** Can we design a system that is robust against sensor failure?
- **Lack of Technical Fluency:** Can we reduce technological complexity?
 - **Repairability:** Need parts that are readily available. Previously, parts were sourced from China.
 - **Maintenance:** Simple, intuitive design for local technicians.
 - **Compatibility:** Should leverage current infrastructure and mobile phone technology.

PROPOSED CHANGES

- Can we improve sensor lifetime? Sensor Redundancy, Housing
- Measurement + Control → Measurement + Recommendation
 - Reduces challenges related to repairability of complex system.
 - Communication: Visual explanations, variety of languages (16 official languages).
- Fuzzy Logic → TinyML Classification Algorithm
- Change the sensor choice to improve microbial indicator performance.



Water Quality Index (WQI)

PROPOSED PLAN

- **Phase A:** Proxy Selection → Data Collation and Dimensional Reduction.
 - **Phase B:** Train TinyML → Gradient Boosting.
 - **Phase C:** Software Design → Constraints: Memory, Connectivity, and Power.
 - **Phase D:** Microcontroller Build → Arduino, sensors, LCD Display.
 - **Phase E:** Second Trial! Testing and data collection of the design in the field.
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PHASE A & B

- Evaluate, clean and collate data from a variety of datasets with differing parameter coverage, reliability, and completeness.
- Use a subset to train a TinyML classification model to determine potability risks.
- Use the remaining data to test the model's ability to generalise across three simulated situations (WNTR):
 1. Contamination event
 2. Slow sensor drift
 3. Chlorine sensor fouling

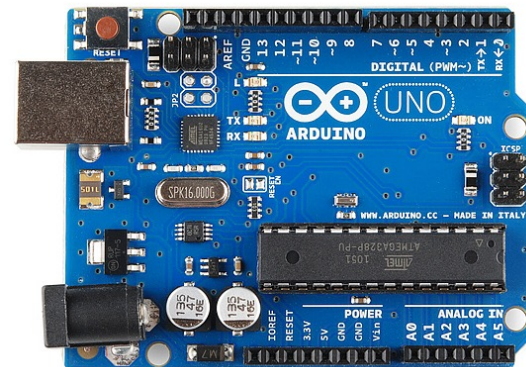


PHASE C

- Design the signal and communication architecture, comparing methods including:
 - Periodically uploading data to a central server.
 - Storing readings locally on-device for later retrieval.
 - Assess trade-offs for connectivity reliability, power consumption, data latency and cost of operation.
 - Implement and test within the memory and power constraints of the microcontroller, ensuring the communication layer integrates with the Phase A output.
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PHASE D

- Construct the microcontroller unit, including Arduino board, selected sensors, and interfacing modules.
- Design and implement sensor protection.
- Run in house testing before graduating to Phase E.



BILL OF MATERIALS (BOM)

Item	Cost (£)
XIAO ESP32C3	14.62
Liquid PH Sensor (Module Accuracy 0.1pH)	35.66
Analog Turbidity Sensor	7.40
Conductivity Sensor	9.01
0.96" OLED Display Module (128x64)	4.00
Digital RGB LED	1.42
3.7 V, Lithium Polymer Rechargeable Battery	14.76
Lipo Charger	4.41
Plastic Project Box	1.44
Total	92.72

FORECASTING CHALLENGES

- **Datasets:**
 - Poor generalization.
 - Incomplete/faulty datasets.
 - **UX:** Ensuring the sensors are appropriately placed (and for a sufficient amount of time).
 - **Motivation and Consistency:** Requires autonomous humans to take responsible actions.
 - **Cost:** Optimal sensor requirement may outprice market WTP.
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QUESTIONS?